

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1411

Stepstone 100 kV
December 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1411. The Interconnection Customer proposes a 75.9 MW battery energy storage facility interconnecting at the Stepstone 100 kV substation in the Indiana area. The request was received on February 22, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 556 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$373,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Scenic Woods to Twin Branch Pv 100 KV	Kemper County to Twin Branch 100 KV L	132.8	170	11.6
High Falls - Croghan Ny to Scenic Woods 100 KV line	Kemper County to Twin Branch 100 KV line	127.9	143	11.6
Denver Terminal to N Mesquite 100 KV	Scenic Woods to Twin Branch Pv 100 KV L	118.3	144	8.4
Parkrose to Twin Branch Pv 100 KV	Twin Branch to Kemper County 100 KV L	107.2	191	11.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Scenic Woods to Twin Branch Pv 100 KV	Rebuild line with higher-capacity conductor	\$2,237,000	4% (\$98,000)
High Falls - Croghan Ny to Scenic Woods 100 KV line	Rebuild line with higher-capacity conductor	\$2,432,000	4% (\$104,000)
Denver Terminal to N Mesquite 100 KV	Rebuild line with higher-capacity conductor	\$1,761,000	4% (\$76,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parkrose to Twin Branch Pv 100 KV	Reconductor line	\$2,171,000	4% (\$95,000)
	Total allocated to project		\$373,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.